

Compute interim envy-free lotteries maximizing expected average Nash welfare in matching instances

ChatGPT (gpt-5.6-sol)*

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Abstract

We study lotteries over perfect assignments of indivisible goods subject to interim envy-freeness. The objective is the expectation of the average Nash welfare of the realized assignment. We prove that maximizing this objective is NP-hard, even when all valuations are strictly positive and belong to the fixed set

$$\left\{ \frac{1}{4}, 1, \frac{7}{4}, 2, \frac{5}{2} \right\},$$

and even when the existence of an interim-envy-free lottery is guaranteed.

The reduction is from INDEPENDENT SET. For every graph G , we construct an instance whose optimum is an explicit strictly increasing function of the independence number $\alpha(G)$. We also give the exponential-size linear-program formulation, prove an $O(n^3)$ support bound, and identify a consistency obstruction in formulations based only on ordered pairs of agents and goods.

1 The open problem

This paper addresses an open problem posed by Ioannis Caragiannis, Panagiotis Kanellopoulos, Maria Kyropoulou in *On Interim Envy-Free Allocation Lotteries* [1] (Mathematics of Operations Research, 2025), Page 19, Section 7 (problems). The website catalogue entry is problem 61 (W3133568468_p0) of the [Open Problems in Operations Research](#) collection.

Given a matching allocation instance $(N, I, (v_i)_{i \in N})$ with $|N| = |I|$, determine how to compute an interim envy-free lottery Q that maximizes the expected average Nash social welfare

$$\max_{Q \text{ iEF}} \mathbb{E}_{\mu \sim Q} [\text{avN}(\mu)].$$

In particular, design an algorithm for this optimization problem (or determine its computational complexity).

*Generated by the `base_solver_non_interactive` (MOR sweep) pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.9 write-up; run `run_20260821_183310_76be8d2e, sweep-audit-approved`). Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

2 Introduction

Let n agents be assigned n indivisible goods, one good per agent. A lottery over assignments is interim envy-free if, after an agent learns which good she receives, she does not prefer the conditional distribution of the good received by any other agent.

The average Nash welfare of a deterministic assignment μ is

$$\text{avN}(\mu) = \left(\prod_{i=1}^n v_i(\mu(i)) \right)^{1/n}.$$

We consider the objective

$$\mathbb{E}_{\mu \sim x}[\text{avN}(\mu)],$$

where x is an interim-envy-free lottery. Note that the expectation is taken outside the geometric mean; this differs from taking the Nash welfare of the agents' ex ante expected utilities.

Our main result is an exact correspondence between the optimum of this problem and the maximum independent-set size of a graph. In particular, if G has m vertices, the constructed instance satisfies

$$\text{OPT}(G) = \left(2^{m-\alpha(G)} \left(\frac{5}{2} \right)^{\alpha(G)} \right)^{1/(7m)}.$$

The result is computational: it does not address extremal bounds on the price of interim envy-freeness.

3 Model

Let $A = [n]$ be the set of agents and let H be a set of n goods. Every agent i has a strictly positive rational value $v_i(g)$ for each good $g \in H$.

A deterministic assignment is a bijection

$$\mu : A \longrightarrow H.$$

Let $\mathcal{M}$ denote the set of all such assignments. A lottery is a vector $x = (x_\mu)_{\mu \in \mathcal{M}}$ satisfying

$$x_\mu \geq 0, \quad \sum_{\mu \in \mathcal{M}} x_\mu = 1.$$

For an agent i and a good g , define the marginal probability

$$p_{ig} = \Pr_{\mu \sim x}[\mu(i) = g] = \sum_{\mu: \mu(i)=g} x_\mu.$$

Definition 1 (Interim envy-freeness). A lottery x is *interim envy-free*, abbreviated iEF, if for every pair of distinct agents i, j and every good g such that $p_{ig} > 0$,

$$v_i(g) \geq \mathbb{E}_{\mu \sim x} [v_i(\mu(j)) \mid \mu(i) = g].$$

Multiplying by p_{ig} shows that this condition is equivalent to

$$\sum_{\mu: \mu(i)=g} x_{\mu} (v_i(g) - v_i(\mu(j))) \geq 0. \quad (1)$$

If $p_{ig} = 0$, both sides of the multiplied condition are zero, so no conditional requirement is imposed.

For a deterministic assignment μ , define

$$w_{\mu} = \text{avN}(\mu) = \left(\prod_{i=1}^n v_i(\mu(i)) \right)^{1/n}.$$

The expected average Nash welfare of a lottery x is

$$\text{EavN}(x) = \sum_{\mu \in \mathcal{M}} x_{\mu} w_{\mu}.$$

4 The outcome linear program

The optimization problem admits the following direct linear formulation:

$$\begin{aligned} \max_x \quad & \sum_{\mu \in \mathcal{M}} w_{\mu} x_{\mu}, \\ \text{subject to} \quad & \sum_{\mu: \mu(i)=g} x_{\mu} (v_i(g) - v_i(\mu(j))) \geq 0 \quad (i \neq j, g \in H), \\ & \sum_{\mu \in \mathcal{M}} x_{\mu} = 1, \\ & x_{\mu} \geq 0 \quad (\mu \in \mathcal{M}). \end{aligned} \quad (\text{LP})$$

There are $n!$ variables and

$$K = n^2(n-1)$$

iEF inequalities.

Proposition 1 (Sparse optimal lotteries). *Whenever the iEF polytope is nonempty, every linear objective over that polytope has an optimal lottery supported on at most*

$$K + 1 = n^2(n-1) + 1$$

assignments. In particular, the expected average Nash welfare admits an optimal lottery with $O(n^3)$ support.

Proof. The feasible region of (LP) is a compact polytope, so a linear objective attains its optimum at a vertex.

Suppose a vertex x has s positive coordinates. Apart from the nonnegativity constraints, at most K iEF inequalities and the normalization equality can be active. If $s > K + 1$, the columns corresponding to the positive coordinates admit a nonzero direction d satisfying the normalization equality and every active iEF equality. For sufficiently small $\varepsilon > 0$, both $x + \varepsilon d$ and $x - \varepsilon d$ remain nonnegative and satisfy all inactive inequalities as well. This contradicts extremality. Hence $s \leq K + 1$. $\square$

Remark (Exact and approximate computation). When the valuations are rational, the feasible region of (LP) is rational, but the objective coefficients

$$w_\mu = \left(\prod_i v_i(\mu(i)) \right)^{1/n}$$

are generally algebraic rather than rational. Therefore, an exact solution requires algebraic-number comparisons or an equivalent real-algebraic decision procedure; an ordinary rational LP solver alone does not provide an exact comparison of objective values.

For additive approximation, suppose rational coefficients $\tilde{w}_\mu$ satisfy

$$|w_\mu - \tilde{w}_\mu| \leq \delta \quad \text{for every } \mu.$$

If $x^\star$ is optimal for w and $\tilde{x}$ is optimal for $\tilde{w}$, then

$$\begin{aligned} w \cdot x^\star - w \cdot \tilde{x} &\leq (w - \tilde{w}) \cdot x^\star + (\tilde{w} - w) \cdot \tilde{x} \\ &\leq 2\delta. \end{aligned}$$

Thus coefficient accuracy $\delta = \varepsilon/2$ suffices for an additive- ε solution.

We next record a necessary condition that will enforce the structure of the hardness construction.

Lemma 1 (Row-average rule). *Let x be an iEF lottery. If $p_{ig} > 0$, then*

$$n v_i(g) \geq \sum_{h \in H} v_i(h).$$

Equivalently, an agent can receive with positive probability only a good whose value is at least the average of that agent's valuation row.

Proof. Fix i and g , and sum (1) over all $j \neq i$. This gives

$$\begin{aligned} 0 &\leq \sum_{j \neq i} \sum_{\mu: \mu(i)=g} x_\mu (v_i(g) - v_i(\mu(j))) \\ &= p_{ig} \left((n-1)v_i(g) - \sum_{h \neq g} v_i(h) \right). \end{aligned}$$

The equality holds because, whenever $\mu(i) = g$, the other agents receive every good in $H \setminus \{g\}$ exactly once. Since $p_{ig} > 0$,

$$(n-1)v_i(g) \geq \sum_{h \neq g} v_i(h),$$

which rearranges to the claimed inequality. $\square$

5 Reduction from Independent Set

Let $G = (V, E)$ be a simple undirected graph with

$$V = [m].$$

Orient every edge from its lower-numbered endpoint to its higher-numbered endpoint, and write

$$N_i^+ = \{j > i : \{i, j\} \in E\}, \quad d_i^+ = |N_i^+|.$$

For each vertex i , create seven agents

$$s_i, \quad r_i, \quad z_i^1, z_i^2, z_i^3, \quad u_i^1, u_i^2$$

and seven goods

$$a_i, \quad b_i, \quad h_i, \quad c_i^1, c_i^2, \quad d_i^1, d_i^2.$$

We refer to these agents and goods as block i . The resulting instance has

$$n = 7m$$

agents and goods.

Every valuation not explicitly listed below equals $1/4$. The nondefault values are

Agent	Nondefault valuations	
s_i	$v_{s_i}(a_i) = 2, \quad v_{s_i}(b_i) = 1,$ $v_{s_i}(a_j) = 1, \quad v_{s_i}(b_j) = \frac{7}{4} \quad \text{for every } j \in N_i^+,$	(2)
r_i	$v_{r_i}(h_i) = \frac{5}{2}, \quad v_{r_i}(d_i^1) = 1,$	
$z_i^q, \quad q \in \{1, 2, 3\}$	$v_{z_i^q}(a_i) = v_{z_i^q}(h_i) = v_{z_i^q}(c_i^1) = v_{z_i^q}(c_i^2) = 1,$	
$u_i^q, \quad q \in \{1, 2\}$	$v_{u_i^q}(b_i) = v_{u_i^q}(d_i^1) = v_{u_i^q}(d_i^2) = 1.$	

Thus the raw valuation alphabet is

$$\left\{ \frac{1}{4}, 1, \frac{7}{4}, 2, \frac{5}{2} \right\}.$$

5.1 Eliminating default entries

Relative to the all- $1/4$ baseline, the excess row sums are

$$\begin{aligned}
s_i : \quad & (2 - \frac{1}{4}) + (1 - \frac{1}{4}) + d_i^+ \left[(1 - \frac{1}{4}) + (\frac{7}{4} - \frac{1}{4}) \right] = \frac{5}{2} + \frac{9}{4}d_i^+, \\
r_i : \quad & (\frac{5}{2} - \frac{1}{4}) + (1 - \frac{1}{4}) = 3, \\
z_i^q : \quad & 4(1 - \frac{1}{4}) = 3, \\
u_i^q : \quad & 3(1 - \frac{1}{4}) = \frac{9}{4}.
\end{aligned}$$

Consequently, the row averages are

$$\bar{v}_{s_i} = \frac{1}{4} + \frac{\frac{5}{2} + \frac{9}{4}d_i^+}{7m}, \quad (3)$$

$$\bar{v}_{r_i} = \bar{v}_{z_i^q} = \frac{1}{4} + \frac{3}{7m}, \quad (4)$$

$$\bar{v}_{u_i^q} = \frac{1}{4} + \frac{9}{28m}. \quad (5)$$

All of these quantities are strictly larger than $1/4$ and strictly smaller than 1 . For example, since $d_i^+ \leq m - 1$,

$$\bar{v}_{s_i} \leq \frac{1}{4} + \frac{9m + 1}{28m} = \frac{16m + 1}{28m} < 1.$$

It follows from Lemma 1 that no agent can receive a default-value good in any positive-probability assignment of an iEF lottery. Hence every assigned agent–good pair in such an assignment must be one of the nondefault pairs in (2).

Note that $9/4$ appears in (3)–(5) only as an aggregate excess; it is not a valuation entry.

5.2 The two block modes

Lemma 2 (Structural lemma). *Every assignment in the positive support of an iEF lottery is block-local. Moreover, each block is in exactly one of the following two modes:*

	<i>unselected mode</i>	<i>selected mode</i>
s_i	a_i	b_i
r_i	d_i^1	h_i
$\{z_i^1, z_i^2, z_i^3\}$	$\{h_i, c_i^1, c_i^2\}$	$\{a_i, c_i^1, c_i^2\}$
$\{u_i^1, u_i^2\}$	$\{b_i, d_i^2\}$	$\{d_i^1, d_i^2\}$

(6)

The braces indicate that the displayed goods may be permuted among the corresponding agents.

Proof. Fix a positive-probability assignment μ . By the row-average rule, all six nonsignal agents

$$r_i, \quad z_i^1, z_i^2, z_i^3, \quad u_i^1, u_i^2$$

must receive nondefault goods. Every nondefault good for these agents belongs to block i , so these six agents receive six distinct goods from their own block.

Exactly one good in block i remains for a signal agent. That good must be either a_i or b_i , because signal agents have default value for all the other local goods.

Define $f(i)$ to be the index of the block containing the good assigned to the signal agent s_i . Every block receives exactly one signal agent, so f is a permutation of $[m]$. By construction, s_i has a nondefault value only for a_i, b_i and for a_j, b_j with $j \in N_i^+$. Therefore

$$f(i) \geq i \quad \text{for every } i.$$

Since f is a permutation,

$$\sum_{i=1}^m f(i) = \sum_{i=1}^m i.$$

Together with $f(i) \geq i$, this forces $f(i) = i$ for every i . Hence every signal agent also remains in its own block.

It remains to determine the possible local assignments.

If s_i receives a_i , the three z_i -agents have only

$$h_i, c_i^1, c_i^2$$

available among their nondefault goods, so they must receive exactly these three goods. Agent r_i can then no longer receive h_i and must receive d_i^1 . The two u_i -agents receive b_i, d_i^2 . This is the unselected mode.

If s_i receives b_i , the two u_i -agents must receive d_i^1, d_i^2 . Agent r_i must therefore receive h_i , and the three z_i -agents receive a_i, c_i^1, c_i^2 . This is the selected mode. $\square$

For a supported assignment μ , define its selected set by

$$S(\mu) = \{i \in [m] : \mu(s_i) = b_i\}.$$

5.3 Interim envy-freeness enforces independence

Lemma 3 (Edge exclusion). *For every assignment μ in the positive support of an iEF lottery, the set $S(\mu)$ is an independent set of G .*

Proof. Let x be an iEF lottery and consider an edge oriented as $i \rightarrow j$. Apply the multiplied iEF constraint (1) for owner s_i , own good b_i , and comparator s_j :

$$0 \leq \sum_{\mu: \mu(s_i)=b_i} x_\mu (v_{s_i}(b_i) - v_{s_i}(\mu(s_j))).$$

By Lemma 2, s_j receives a_j when j is unselected and b_j when j is selected. Therefore

$$v_{s_i}(\mu(s_j)) = \begin{cases} 1, & j \notin S(\mu), \\ 7/4, & j \in S(\mu). \end{cases}$$

Since $v_{s_i}(b_i) = 1$, the constraint becomes

$$0 \leq -\frac{3}{4} \sum_{\mu: i, j \in S(\mu)} x_\mu.$$

Every term in the last sum is nonnegative, so

$$\sum_{\mu: i, j \in S(\mu)} x_\mu = 0.$$

Thus no positive-probability assignment selects both endpoints of the edge. $\square$

6 Implementing every independent set

We now prove the converse: every independent set can be implemented by an iEF lottery.

Lemma 4 (Six-rotation construction). *For every independent set $S \subseteq V$, there is an iEF lottery supported on six assignments such that every supported assignment has selected set exactly S .*

Proof. Fix an independent set S . Put block i in the selected mode when $i \in S$, and in the unselected mode otherwise.

For each block, define its ordered z -basket and u -basket by

	z -basket	u -basket
$i \notin S$	(h_i, c_i^1, c_i^2)	(b_i, d_i^2)
$i \in S$	(a_i, c_i^1, c_i^2)	(d_i^1, d_i^2)

The assignments of s_i and r_i are fixed as in (6).

For $t \in \{0, 1, \dots, 5\}$, define μ_t as follows:

- cyclically shift every z -basket by $t \bmod 3$ among z_i^1, z_i^2, z_i^3 ;
- swap every u -basket according to $t \bmod 2$ between u_i^1, u_i^2 .

The same phase t is used in every block. Let x^S be the uniform lottery over $\mu_0, \dots, \mu_5$.

We now verify interim envy-freeness.

Every z -agent and every u -agent receives utility 1 in every realization, and every entry in each such agent's valuation row is at most 1. Consequently, these agents cannot envy, even after conditioning on a realized good that reveals the global phase.

A selected r_i receives h_i with value $5/2$, its row maximum. An unselected s_i receives a_i with value 2, also its row maximum. These agents cannot envy either.

Consider an unselected r_i . It receives d_i^1 with value 1 in every realization. Its only good worth more than 1 is h_i , which rotates uniformly through the three z_i -agents. For every fixed z_i^q ,

$$\mathbb{E}[v_{r_i}(\mu_t(z_i^q))] = \frac{\frac{5}{2} + \frac{1}{4} + \frac{1}{4}}{3} = 1.$$

All of its other comparisons have value at most 1.

Finally, consider a selected s_i . It receives b_i with value 1. The local good a_i rotates through the three z_i -agents, and hence, for every fixed z_i^q ,

$$\mathbb{E}[v_{s_i}(\mu_t(z_i^q))] = \frac{2 + \frac{1}{4} + \frac{1}{4}}{3} = \frac{5}{6} \leq 1.$$

It remains to check the outgoing neighbors of i . If $i \rightarrow j$, then $j \notin S$ because S is independent. Thus s_j receives a_j , which has value 1 to s_i . The good b_j rotates uniformly between u_j^1, u_j^2 , paired with d_j^2 . Therefore, for either fixed u_j^q ,

$$\mathbb{E}[v_{s_i}(\mu_t(u_j^q))] = \frac{\frac{7}{4} + \frac{1}{4}}{2} = 1.$$

Every other good in block j has value $1/4$ to s_i , and non-neighboring blocks likewise contain only default-value goods from the perspective of s_i .

The agents whose constraints require averaging, namely an unselected r_i and a selected s_i , receive the same good in all six assignments. Conditioning on that good therefore does not alter the uniform distribution over phases. Although the shared phase correlates different blocks, no independence across blocks is needed for any of the preceding calculations. Hence x^S is iEF. $\square$

Taking $S = \emptyset$ shows that every constructed instance admits an iEF lottery.

7 Welfare and the exact optimum

In an unselected block, the agents' utilities are

$$2, 1, 1, 1, 1, 1, 1,$$

so the block product is

$$P_0 = 2.$$

In a selected block, the utilities are

$$1, \frac{5}{2}, 1, 1, 1, 1, 1,$$

so the block product is

$$P_1 = \frac{5}{2}.$$

Consequently, every supported assignment selecting k vertices has average Nash welfare

$$W_k = \left(2^{m-k} \left(\frac{5}{2} \right)^k \right)^{1/(7m)}. \quad (7)$$

The sequence $W_0, \dots, W_m$ is strictly increasing because

$$\frac{W_k}{W_{k-1}} = \left(\frac{5}{4} \right)^{1/(7m)} > 1.$$

Theorem 1 (Exact correspondence). *Let $\mathcal{I}(G)$ be the allocation instance constructed from G . Then*

$$\text{OPT}(\mathcal{I}(G)) = W_{\alpha(G)} = \left(2^{m-\alpha(G)} \left(\frac{5}{2} \right)^{\alpha(G)} \right)^{1/(7m)}.$$

Proof. Let x be any iEF lottery. By Lemma 3, every assignment $\mu \in \text{supp}(x)$ selects an independent set. Hence

$$|S(\mu)| \leq \alpha(G).$$

Since W_k is increasing,

$$\text{avN}(\mu) = W_{|S(\mu)|} \leq W_{\alpha(G)}.$$

Averaging over μ gives

$$\text{EavN}(x) \leq W_{\alpha(G)}.$$

Conversely, let S be a maximum independent set. By Lemma 4, there is an iEF lottery supported on six assignments, each having selected set S . Every such assignment has welfare $W_{\alpha(G)}$, so the lottery attains the upper bound. $\square$

8 A rational-threshold reduction

The exact values W_k are algebraic. We next show that the decision problem remains NP-hard when the threshold is required to be rational.

Lemma 5 (Inverse-polynomial separation). *For every $k \in \{1, \dots, m\}$,*

$$W_k - W_{k-1} > \frac{1}{35m}.$$

Proof. Let

$$\theta = \frac{\ln(5/4)}{7m}.$$

Then

$$W_k - W_{k-1} = W_{k-1}(e^\theta - 1).$$

Since $W_{k-1} > 1$ and $e^\theta - 1 > \theta$,

$$W_k - W_{k-1} > \frac{\ln(5/4)}{7m}.$$

Moreover,

$$\ln\left(\frac{5}{4}\right) = \int_1^{5/4} \frac{dt}{t} > \frac{1}{4} \cdot \frac{4}{5} = \frac{1}{5}.$$

The result follows. □

Theorem 2 (NP-hardness). *The following decision problem is NP-hard:*

Given strictly positive rational valuations and a rational threshold T , is there an iEF lottery whose expected average Nash welfare is at least T ?

The result holds even when all valuation entries belong to

$$\left\{\frac{1}{4}, 1, \frac{7}{4}, 2, \frac{5}{2}\right\}$$

and the existence of an iEF lottery is guaranteed.

Proof. We reduce from INDEPENDENT SET. Given a graph G on m vertices and an integer $k \in \{1, \dots, m\}$, construct $\mathcal{I}(G)$.

Let

$$Q = 36m, \quad r = \lfloor QW_{k-1} \rfloor, \quad T = \frac{r+1}{Q}.$$

Then

$$0 < T - W_{k-1} \leq \frac{1}{Q} = \frac{1}{36m}.$$

By Lemma 5,

$$W_{k-1} < T < W_k.$$

The integer r can be computed in polynomial time using exact rational arithmetic. Indeed, for a candidate integer ℓ , the comparison

$$\frac{\ell}{Q} \leq W_{k-1}$$

is equivalent to

$$\left(\frac{\ell}{Q}\right)^{7m} \leq 2^{m-k+1} \left(\frac{5}{2}\right)^{k-1}.$$

After clearing denominators, this is a comparison of integers with polynomially many bits. Binary search therefore computes r , and T has polynomial encoding length.

By Theorem 1,

$$\begin{aligned} \alpha(G) \geq k &\implies \text{OPT}(\mathcal{I}(G)) \geq W_k > T, \\ \alpha(G) \leq k-1 &\implies \text{OPT}(\mathcal{I}(G)) \leq W_{k-1} < T. \end{aligned}$$

Thus

$$\alpha(G) \geq k \iff \text{OPT}(\mathcal{I}(G)) \geq T.$$

The construction is polynomial, uses only the displayed fixed valuation alphabet, and is always feasible by Lemma 4 with $S = \emptyset$. $\square$

9 Consequences for other welfare objectives

The same construction also records the selected-set size in both log-Nash and utilitarian welfare.

Let

$$\lg N(\mu) = \sum_i \ln v_i(\mu(i)), \quad U(\mu) = \sum_i v_i(\mu(i)).$$

Normalizing $\lg N$ by $7m$ would not affect its maximizers.

Corollary 1. *For the constructed instance,*

$$\max_{x \in iEF} \mathbb{E}_x[\lg N(\mu)] = m \ln 2 + \alpha(G) \ln \left(\frac{5}{4}\right)$$

and

$$\max_{x \in iEF} \mathbb{E}_x[U(\mu)] = 8m + \frac{\alpha(G)}{2}.$$

Consequently, exact optimization of either objective over iEF lotteries is NP-hard.

Proof. An unselected block has product 2 and total utility

$$2 + 1 + 1 + 1 + 1 + 1 + 1 = 8.$$

A selected block has product $5/2$ and total utility

$$1 + \frac{5}{2} + 1 + 1 + 1 + 1 + 1 = \frac{17}{2}.$$

Thus an assignment selecting k vertices has

$$\lg N(\mu) = m \ln 2 + k \ln \left(\frac{5}{4}\right)$$

and

$$U(\mu) = 8m + \frac{k}{2}.$$

Every supported selected set is independent, and every independent set is implemented by Lemma 4. The formulas follow. $\square$

10 Why ordered-pair variables are insufficient

The hardness result may appear to conflict with approaches based on variables indexed only by ordered pairs of agents and ordered pairs of goods. We give a direct consistency obstruction.

Let

$$A^{(2)} = \{(i, j) : i, j \in A, i \neq j\}, \quad H^{(2)} = \{(g, h) : g, h \in H, g \neq h\}.$$

Consider variables $y_{ij,gh}$ subject to

$$\sum_{(g,h) \in H^{(2)}} y_{ij,gh} = 1 \quad ((i, j) \in A^{(2)}), \quad (8)$$

$$\sum_{(i,j) \in A^{(2)}} y_{ij,gh} = 1 \quad ((g, h) \in H^{(2)}), \quad (9)$$

$$y_{ij,gh} = y_{ji,hg}, \quad (10)$$

$$y_{ij,gh} \geq 0. \quad (11)$$

Every matching induces such a point, but the converse fails.

Proposition 2 (Pairwise consistency does not imply a matching). *There is a 0-1 extreme point satisfying (8)–(11) that is not induced by any matching or any lottery over matchings.*

Proof. Take three agents 1, 2, 3 and three goods a, b, c . Define a bijection from ordered agent pairs to ordered good pairs by

$$\begin{aligned} (1, 2) &\mapsto (a, b), & (1, 3) &\mapsto (c, a), & (2, 3) &\mapsto (b, c), \\ (2, 1) &\mapsto (b, a), & (3, 1) &\mapsto (a, c), & (3, 2) &\mapsto (c, b). \end{aligned}$$

Set $y_{ij,gh} = 1$ for the six displayed mappings and set all other entries to zero.

Every ordered agent-pair row contains exactly one 1, and every ordered good-pair column contains exactly one 1. The last three mappings are the reversals of the first three, so (10) also holds.

The point cannot be induced by a matching: the mapping

$$(1, 2) \mapsto (a, b)$$

requires agent 1 to receive a , whereas

$$(1, 3) \mapsto (c, a)$$

requires the same agent 1 to receive c .

Nor can the point be the pair marginal of a lottery. Since $y_{12,ab} = 1$, every matching in such a lottery would have to assign a to agent 1. Since $y_{13,ca} = 1$, every matching would simultaneously have to assign c to agent 1, which is impossible.

Finally, the displayed matrix is a permutation matrix and hence a vertex of the full doubly stochastic polytope. Since it also satisfies the reversal constraint, it remains a vertex of the reversal-symmetric subpolytope. $\square$

A characterization of the extreme points as reversal-symmetric or centrosymmetric permutation matrices establishes only that an extreme point is a bijection

$$\Phi : A^{(2)} \longrightarrow H^{(2)}$$

satisfying

$$\Phi(j, i) = \text{rev}(\Phi(i, j)).$$

It does not establish the much stronger factorization

$$\Phi(i, j) = (\mu(i), \mu(j))$$

for a single underlying matching μ . That missing global consistency condition prevents the ordered-pair formulation from being a valid replacement for the outcome formulation (LP).

11 Conclusion

Maximizing expected average Nash welfare over interim-envy-free lotteries is NP-hard even on feasible instances with a fixed, strictly positive valuation alphabet. The reduction gives the exact identity

$$\text{OPT}(G) = \left(2^{m-\alpha(G)} \left(\frac{5}{2} \right)^{\alpha(G)} \right)^{1/(7m)}.$$

The six-rotation construction proves that every independent set is attainable, while the interim constraint on signal agents ensures that only independent sets can occur in positive support.

The outcome LP nevertheless gives an exponential formulation and an $O(n^3)$ optimal-support bound. Any purported polynomial-time replacement based only on ordered pairs must additionally prove global extendability to a common lottery; pairwise row, column, and reversal constraints alone do not provide it.

References

- [1] Ioannis Caragiannis, Panagiotis Kanellopoulos, Maria Kyropoulou. *On Interim Envy-Free Allocation Lotteries*. Mathematics of Operations Research, 2025. <https://doi.org/10.1287/moor.2023.0203>